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Chapter 1

Standard Model of Cosmology

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Introduction

Cosmology is the study of the Universe, aiming to understand its origin and evolution. Within the broader landscape of science and physics, it occupies a unique position: it is one of the few fields where phenomena cannot be reproduced to directly test theoretical predictions. We have access to a single realization of the Universe, cannot replicate its underlying mechanisms, and can only observe a limited portion of its information.

This thesis aims to contribute to our understanding of the earliest moments of the Universe. This first chapter provides a structured overview of the standard cosmological model, known as ΛCDM, guiding the reader from its fundamental principles to its current limitations. The chapter is organized into five sections. The first introduces the key components of the model, including its foundations, assumptions, and limitations, while the second presents a qualitative overview of the thermal history of the Universe. The third focuses on the Cosmic Microwave Background (CMB), one of the most powerful probes of the early Universe. The fourth section presents the main sources of contamination affecting CMB observations. Finally, the fifth section reviews past CMB experiments and their major results, before discussing upcoming missions.

1.1 The Standard Model of Cosmology

Modern cosmology is built upon a set of simple but powerful principles that allow us to describe the large-scale behaviour of the Universe. Observations reveal that the Universe is expanding and appears remarkably homogeneous and isotropic on large scales. These properties provide the foundation for the standard cosmological model, known as ΛCDM.

In this section, we introduce the key elements of this framework. We begin with the observational evidence for the expansion of the Universe, then present the assumptions underlying its large-scale description. We subsequently describe the main components of the Universe, before discussing the limitations of this model and the role of inflation as a possible resolution.

1.1.1 Expanding universe

A major milestone in modern cosmology was the discovery that the Universe is expanding, independently established by Lemaître in 1927 [1] and Hubble in 1929 [2]. Their observations showed that distant galaxies recede from each other, with a velocity proportional to their distance. This relation, now known as Hubble-Lemaître law, provided the first evidence that the Universe is not static but evolving.

This expansion is described by the *scale factor* $a(t)$, which quantifies how distances between comoving points evolve with time. By convention, the scale factor is normalized to $a(t_0) = 1$ today. As the Universe expands, the scale factor increases, implying that physical distances between distant, gravitationally unbound objects grow proportionally with time, as illustrated in Figure 1.1.

An observable consequence of this expansion is the *cosmological redshift* z. As light propagates through an expanding Universe, its wavelength is stretched along with space. This effect is not a simple Doppler redshift [3], but rather a consequence of the expansion of spacetime, hence the name of cosmological redshift, illustrated in Figure 1.2.

The redshift is defined as:

$$1 + z \equiv \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}} \equiv \frac{a_{\text{obs}}}{a_{\text{emit}}} = \frac{1}{a_{\text{emit}}} \tag{1.1}$$

This relation allows us to infer distances and the expansion history of the Universe from observations of spectral lines. In his original work, Hubble measured the redshift of galaxies and compared it to their distances, establishing the linear relation between velocity and distance shown in Figure 1.3. This observation demonstrates that the expansion of the Universe is uniform on large scales: every observer sees distant galaxies receding, without implying a special position in space.

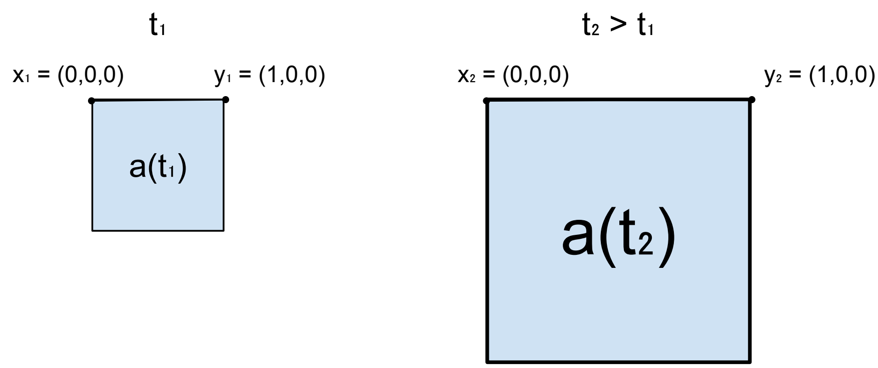

Figure 1.1: Illustration showing the difference between comoving and physical distances. At a time t_1, two points x_1 and y_1 are separated by a distance 1. After some time, at t_2, the Universe has expanded: the scale factor increased and the physical distance between x and y is larger proportionally to the scale factor. In comoving coordinates, however, the distance between x_2 and y_2 remains unchanged.

1.1.2 Cosmological principle

The fundamental concept of the cosmological principle states that the universe is homogeneous and isotropic at large scales. This idea was introduced as a simplification to solve the general relativity equations [5]. It was done by Einstein in 1917 in the case of a static universe [6], but as we saw in previous subsection 1.1.1, it was not relevant any more to assume a static universe after Lemaître and Hubble's works. In 1922, Friedmann showed that a non-static solution for a homogeneous universe is possible [7], quickly followed by evidences of universe expansion in 1927 by Lemaître [1]. Building on these results, Robertson [8] and Walker [9] formulated the general relativistic metric for a homogeneous and isotropic expanding universe. This metric forms the cornerstone of the ΛCDM model. The veracity of this principle is much debated, but latest result from Planck Satellite suggest that the universe is homogeneous and isotropic at 10^{-5} ($= 0.001\%$) at scale above $6°$ on the sky [10].

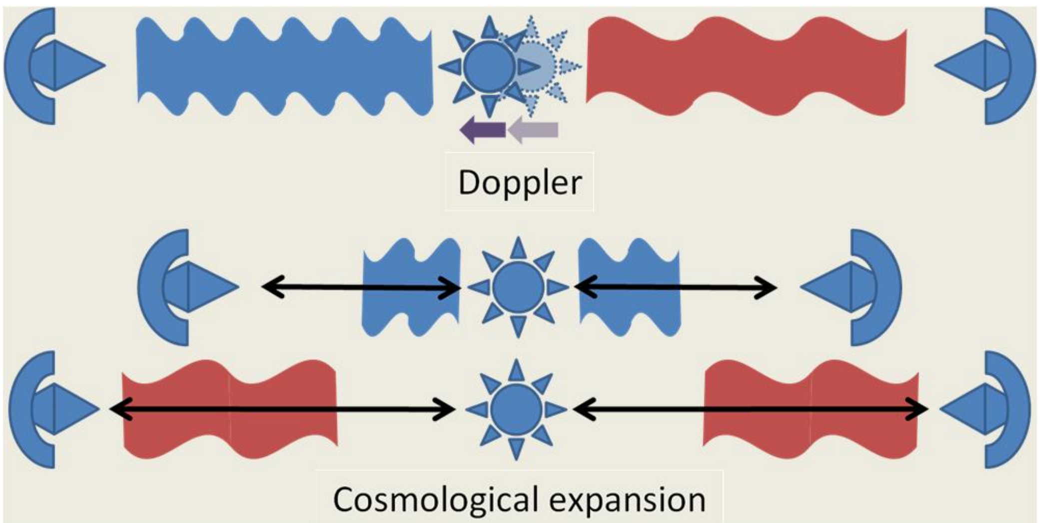

Figure 1.2: Illustration, from [4], showing the difference between Doppler redshift and cosmological redshift. Doppler's redshift [3], on top, is the decrease in wavelength of an object approaching us, and the broadening of an object moving away from us. Cosmological's redshift [2] is the broadening of the wavelength in all directions due to the expansion of the universe.

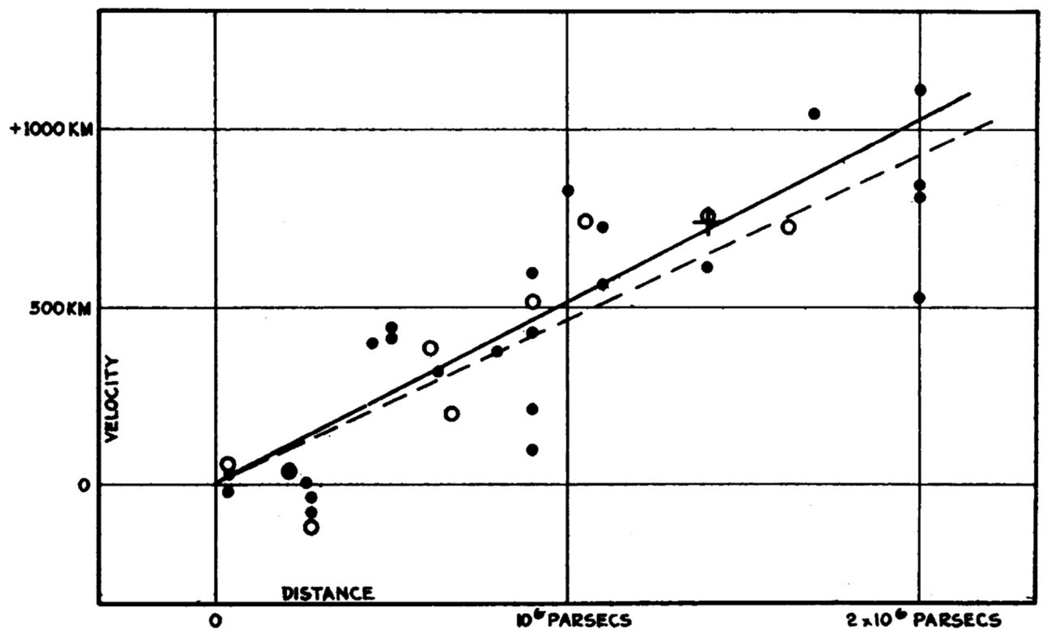

Figure 1.3: Historical velocity-distance diagram from Edwin Hubble [2]. It shows the recession velocity of galaxies as a function of their distance, illustrating the linear relation now known as Hubble's law.

1.1.3 Content of the Universe

In the two previous subsections 1.1.1 and 1.1.2, we introduced the standard assumptions about the universe : it is homogeneous, isotropic and expanding. Now, we now provide a brief overview of its contents. These contents can be classified into three categories: radiation, matter, and the cosmological constant. Each of these components evolves differently with the scale factor and, consequently, with the expansion of the universe. Moreover, each component dominated the universe at different epochs in its history. The evolution of each density is shown in Figure 1.4.

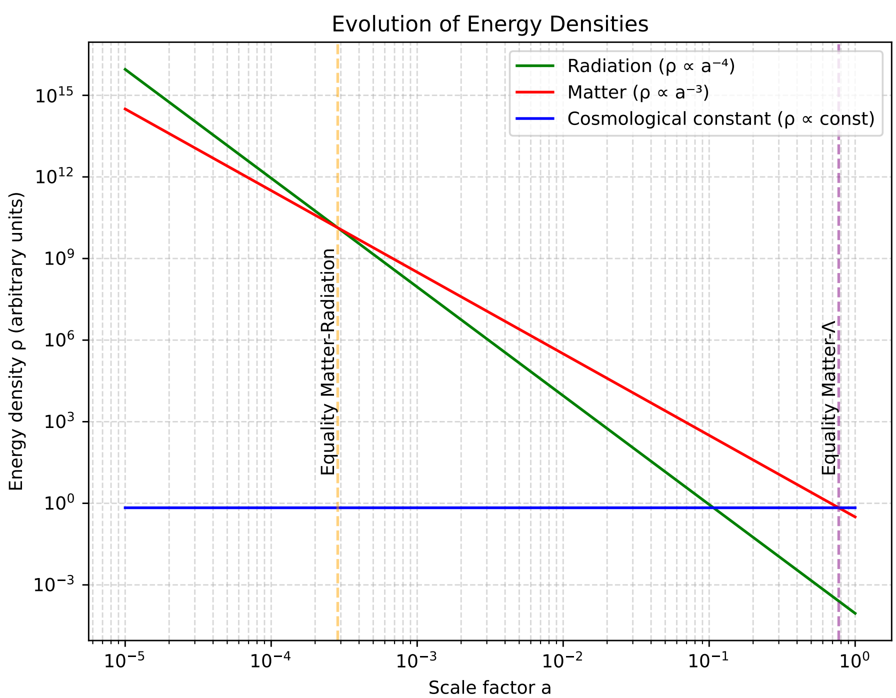

Figure 1.4: Energy density of the Universe's components as a function of the scale factor, highlighting the epochs when matter equals radiation and when matter equals the cosmological constant.

At early times of the universe, the radiation was dominating. This radiation consisted of the relativistic particles : photons and neutrinos, the main contribution being the Cosmic Microwave Background (detailed in Section **??**). The density of radiation and matter were equal at z $\sim$ 3408 according to DESI [11] and z $\sim$ 3387 according to Planck [12], which correspond to approximatively 80000 years after first times of the universe.

Today, radiation contributes about 0.008% of the total energy density, as inferred from Planck 2018 data [12].

As the Universe expanded, non-relativistic matter came to dominate the energy density. This matter is composed of baryonic matter and dark matter. The term "baryonic matter" is misleading, as it designates all non-relativistic particles of the Standard Model; however, its energy density is overwhelmingly dominated by protons and neutrons, hence the terminology. On the other hand, dark matter is a hypothetical form of matter, which only (or mostly) interact through gravitational force, and thus, which does not emit light. It was first suggested by comparing "seen mass" computed from galaxies light and "unseen mass" computed from galaxies velocity in the Coma Galaxies Cluster by Zwicky in 1933 [13]. The mass computed from galaxies velocity was much higher than the luminous mass, suggesting the presence of massive matter that does not emit light.

Since then, multiple independent observations have provided strong evidence for dark matter : galaxy rotation curves [14], the Bullet Cluster through gravitational lensing [15], Baryon Acoustic Oscillations pics observed in CMB [12], formation rate of Large Scale Structures [16], and many others. While we still don't know its nature from the particle point of view, its cosmological properties are well constrained. It is commonly divided into three categories : cold, warm and hot dark matters, which refer to their velocity(or equivalently, whether it was relativistic in the early Universe). This classification determines how far dark matter can propagate (free-stream) and thus influences the formation of structures at different scales. Observations strongly favor the Cold Dark Matter (CDM) scenario [17], in which dark matter is non-relativistic.

Today, Cold Dark Matter (CDM) represents about 26.5% [12] of total energy density, and plays a central role in Standard Model of Cosmology, ΛCDM. The ordinary matter is responsible for 4.5% [12] of the energy density budget, meaning that the energy density of matter occupies 31% of the energy density of the universe, and that 85% of matter in the universe is composes of dark matter.

At z $\sim$ 0.3, corresponding to an age of about 10 Gyr, the Universe transitions from matter domination to dark energy domination, as the cosmological constant becomes the dominant component. The concept of dark energy was first introduced to explain the acceleration of the expansion in the late universe from distant Type Ia supernovae in 1999 [18]. To account for this acceleration, the introduction of a cosmological constant Λ, corresponding to an energy density which does not depend on the scale factor, was needed. Today, dark energy remains one of the greatest unkown in cosmology and in fundamental physics in general.

Dark energy is responsible for 69% of the total energy dansity today, and is by far the main constituent of our universe.

1.1.4 Problems of the Standard Model

From the previous subsection, the Standard Model of cosmology describes the universe as homogeneous, isotropic, expanding, and composed of radiation, matter and dark energy. It is enough to describe the evolution of the universe, and match remarkably well with observations. However, fundamental questions remain unanswered, and in particular three issues cannot be explained within this framework: the horizon problem, the flatness problem, and the monopole problem.

Horizon problem : Why is the CMB temperature so uniform ? As discussed previously, at the time of recombination the Universe was extremely homogeneous, with density fluctuations of order 10^{-5}, and consequently the cosmic microwave background (CMB) exhibits an almost perfectly uniform temperature. This uniformity is naturally explained if the early Universe was in thermal equilibrium, consistent with the fact that the CMB spectrum is an almost perfect blackbody [19]. However, within the standard cosmological model, regions of the CMB that are widely separated on the sky were never in causal contact: light signals would not have had enough time to travel between them since the beginning of the expansion. These regions therefore lie outside each other's particle horizon and cannot have exchanged information or thermalised. This raises a fundamental question: why do causally disconnected regions have the same temperature?

Flatness problem : Why is the universe flat ? In general relativity [5], the geometry of spacetime is determined by its energy content. In a homogeneous and isotropic Universe, this relation is encoded in the Friedmann equations [7], which relate the expansion rate to the total energy density and spatial curvature. From these equations, one defines the critical density, ρ_c, which corresponds to a spatially flat Universe. Depending on the ratio $\Omega = \rho/\rho_c$, the Universe can be:

$$\begin{aligned}
\Omega = 1 &\quad : \text{flat (Euclidean geometry)}, \\
\Omega > 1 &\quad : \text{closed (positive curvature)}, \\
\Omega < 1 &\quad : \text{open (negative curvature)}.
\end{aligned}$$

Current measurements, notably from the Planck mission [12], indicate that Ω differs from unity by less than about one percent, implying that the Universe is extremely close to spatial flatness. However, within the standard Friedmann evolution, flatness is not a stable state: any small deviation from $\Omega = 1$ grows with time. Consequently, evolving backward in time implies an extreme fine-tuning of the initial condition. For example, a deviation of order $|\Omega_0 - 1| \sim 10^{-3}$ today corresponds to an initial deviation of order 10^{-60} near the Planck epoch. This extraordinary sensitivity suggests an apparent fine-tuning problem: why was the early Universe so precisely close to flatness?

Monopole problem : Why are magnetic monopoles not observed ? Grand unified theories (GUTs), which attempt to unify the electromagnetic, weak, and strong interactions at high energy scales, generically predict the existence of topological defects formed during phase transitions in the early Universe. Among these relics are magnetic monopoles, hypothetical particles carrying an isolated magnetic charge, whose existence was first proposed by Dirac in 1931 [20]. In standard cosmology, such monopoles should be efficiently produced during the spontaneous symmetry breaking associated with the GUT phase transition. Causally disconnected regions of space choose different vacuum states, leading to the formation of stable topological defects. As a consequence, one expects a large relic abundance of monopoles. However, monopoles are not observed in the present Universe, despite extensive experimental searches. This discrepancy constitutes the monopole problem: standard cosmology predicts a relic density of monopoles that is many orders of magnitude larger than observational bounds. Should I talk about higher order topological defects ? Like cosmic cords, walls, ...

Finally, an additional issue remains throughout the Standard Model of Cosmology's equations. As with any set of equations describing the evolution of a physical system, the dynamics are only fully specified once appropriate initial conditions are given. This naturally raises the question: how can the initial conditions of our Universe be determined or constrained observationally? An answer to the three problems and to this issue was proposed by Alan Guth in 1981, through the idea of an exponential expansion phase at the very beginning of the Universe [21]. The model was then developed by Andrei Linde in 1983 [22], and has since given rise to a wide variety of models. We will explain in the following subsection the idea behind this theory and how it can solve the issues discussed previously.

1.1.5 Inflation theory

Should I talk about the Inflation scale ($\exp -N$, with $N \sim 60$), to make more understandable that it solves all the problems

The theory of Inflation postulates that in the very early universe, all regions were causally connected. These regions were then disconnected due to an exponentially expanded phase. This theory was developed to solve the various problems presented previously 1.1.4. **Horizon Problem.** Indeed, if all regions were connected in the very early universe, they could have thermalised, explaining why the CMB temperature is uniform at a very large scale. **Flatness Problem.** As we said, a nearly flat universe is an unstable solution, meaning that it is very unlikely to happen. But, Inflation suggests that they do not care about the curvature before it occurs, as an exponential expansion will make all local curvatures negligible. We can understand this argument with the analogy of a sphere: if we zoom enough on a sphere, its surface will be flat. This mechanism explains why the universe is flat today, independently of its initial conditions. **Monopole Problem.** The GUT theories predict the formation of monopoles, but the exponential expansion dilutes their density to the point where they become unobservable today.

Additionally, we talked about all the properties of the universe for the ΛCDM model: homogeneity, isotropy, and flat geometry. These initial conditions seem too "fine-tuned". Inflation also solves this issue by making any specific initial condition negligible due to the exponential expansion. To go even further, Inflation would have expanded quantum fluctuations to a macroscopic scale, responsible for the inhomogeneity of density observed in the temperature fluctuations in the CMB, which led to the large-scale structures' formation under gravitational effects today.

The most widely studied inflationary scenario involves a scalar field that slowly evolves toward its true ground state [23] [24]. In this framework, the field's potential energy remains nearly constant and therefore rapidly comes to dominate over both its kinetic energy and the energy density of other components. Inflation comes to an end when the field reaches the minimum of its potential, after which it undergoes oscillations and decays into lighter particles, eventually producing those of the Standard Model of Particles. A wide variety of potential shapes have been proposed in the literature (see [25] for an overview of the different models), and many of these can be adjusted to fit current observations. This highlights the importance of observational constraints in discriminating between competing inflationary models.

Is it enough on Inflation ? Or should I talk more about the mechanism, slow-roll approximation, ...?

1.2 Thermal history of the universe

In this section, we aim to produce a quick overview of the history of the universe, discussing the key milestones of its evolution and their implication.

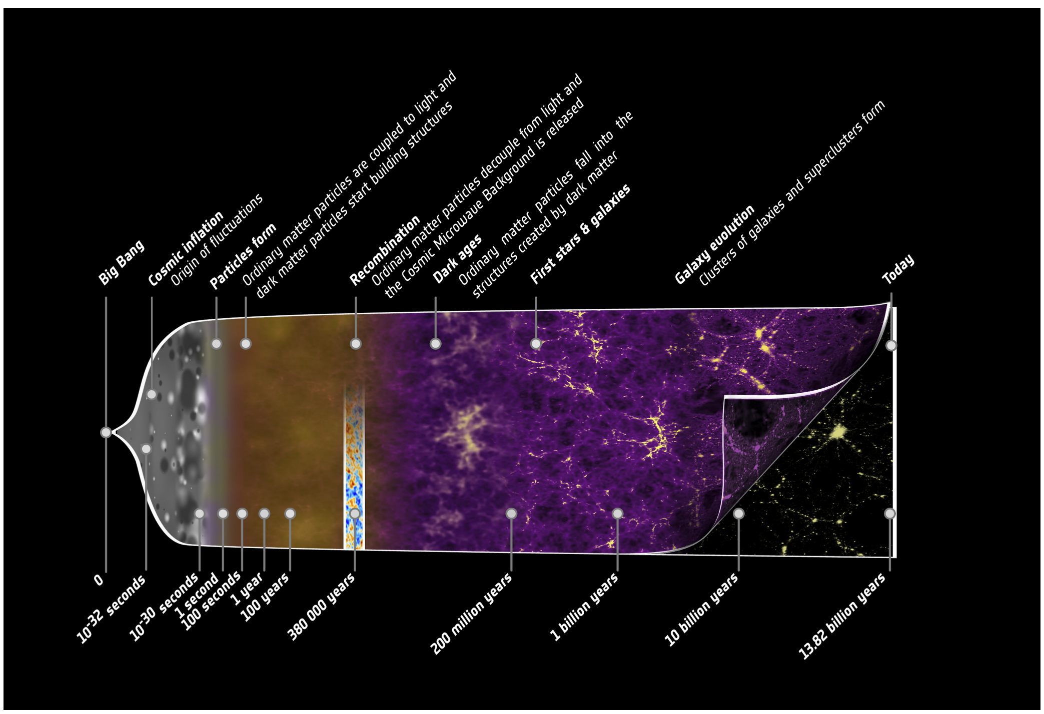

Figure 1.5: Illustration of the history of the universe from the early beginning to today, highlighting the key events discuss in this section. Credit: ESA – C. Carreau.

1.2.1 Inflation and First particles

Is it useful ? Should I add temporal and/or energetic scale(s) ?

As discussed in Section 1.1.5, the standard cosmological model predicts an early phase of accelerated, quasi-exponential expansion known as Inflation. This paradigm addresses several major issues in cosmology and provides a mechanism for setting the initial conditions of the observable Universe. At the end of inflation, the scalar field driving this expansion reached its minimum potential and decays into lighter particles as the universe cool down. This process, known as reheating, initiates the production of the particle content of the Standard Model through a cascade of interactions and decays. Subsequently, matter and antimatter are generated during baryogenesis, with a small asymmetry in favor of matter. Their annihilation produces numerous photons, leaving behind a residual matter component due to this asymmetry. As the temperature continues to decrease, the fundamental interactions—strong, weak, and electromagnetic—become distinct through a sequence of symmetry-breaking transitions. This leads to the formation of hadrons, primarily protons and neutrons. Neutrinos eventually decouple from the primordial plasma, forming a relic background known as the *cosmic neutrino background*. Finally, universe's temperature becomes too low for electron/position pairs production. The remainings electron and positron annihilate to produce photons. The result is a homogeneous universe dominated by radiation (photons, neutrinos) [26].

Is citation OK ? I just want to cite a general reference for this but don't know how to do

1.2.2 Big Bang Nucleosynthesis

At early times, the temperature of the Universe was too high for nuclei to form, as any bound state was immediately broken by high-energy photons. As the Universe expanded and cooled, this process became inefficient, allowing nuclear reactions to proceed.

Weak interactions froze out at a temperature of $T \sim 0.8\,\mathrm{MeV}$ ($t \sim 1\,\mathrm{s}$), fixing the neutron-to-proton ratio at $n/p \sim 1/6$, which subsequently decreased due to free neutron decay. Big Bang Nucleosynthesis (BBN), first introduced in [27], effectively began when the temperature dropped below

$T \sim 0.1\,\text{MeV}$ ($t \sim 200\,\text{s}$), enabling the formation of deuterium through

$$p + n \to {}^2\text{H} + \gamma\,. \tag{1.2}$$

This marked the end of the so-called *deuterium bottleneck*. Rapid nuclear reactions then led to the synthesis of light elements, primarily ^{3}He, ^{4}He, and trace amounts of ^{7}Li. Due to its large binding energy, ^{4}He is the most stable and abundant product after hydrogen, with a final mass fraction of about 25%, while the remaining baryonic matter consists mostly of hydrogen.

The predicted primordial abundances are in good agreement with observations, making BBN one of the key pillars of the standard cosmological model [28]. The Figure 1.6 shows the evolution of abundances of the lighter elements with time and energy.

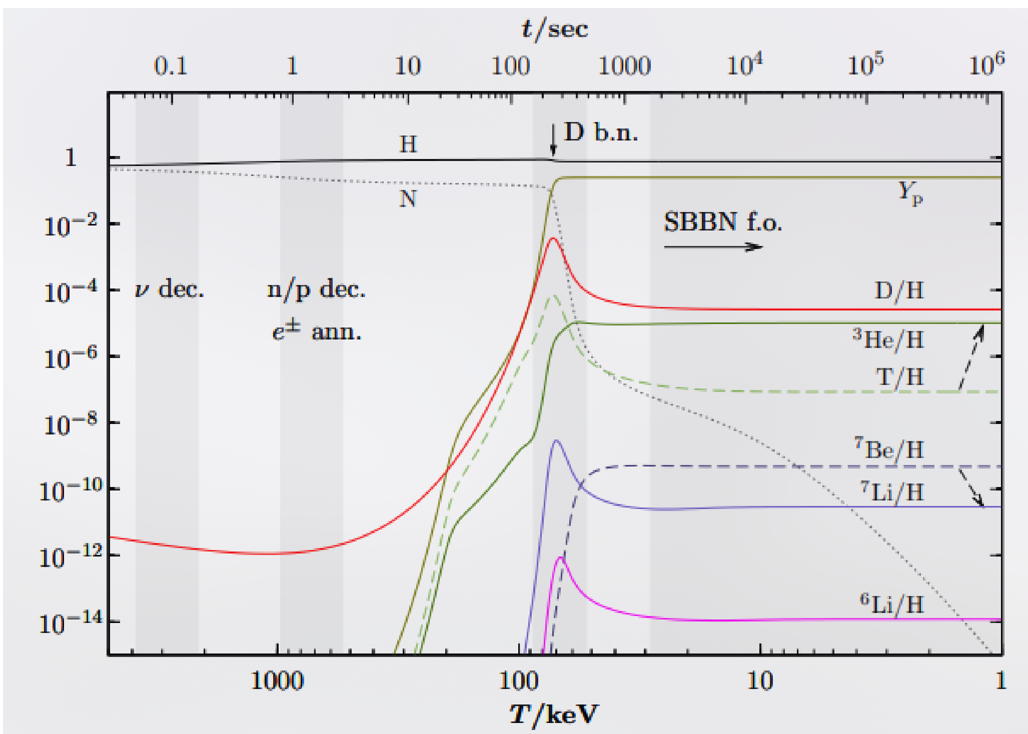

Figure 1.6: Illustration of the Big Bang Nucleosynthesis process, showing the abundances of elements with time (top-axis) and energy (bottom-axis). Taken from [29].

1.2.3 Baryon Acoustic Oscillations

During the radiation-dominated era, photons were tightly coupled to baryons through Thomson scattering of free electrons [26], forming a single photon-baryon fluid. Due to the primordial density fluctuations generated during inflation (see Section 1.1.5), overdense regions acted as gravitational potential wells, attracting baryonic matter. However, photon pressure counteracted gravitational fall. The competition between gravity and radiation pressure led to acoustic oscillations in the coupled photon-baryon fluid, generating spherical sound waves propagating outward from initial overdensities. These oscillations ceased at recombination (see Section 1.2.4), when photons decoupled from baryons and began to free-stream. The maximum distance traveled by these sound waves defines a characteristic scale, known as the sound horizon. This scale is imprinted in the late-time matter distribution as a preferred clustering scale. Baryon Acoustic Oscillations (BAO) were first detected in the large-scale distribution of galaxies by the Sloan Digital Sky Survey [30] and independently confirmed by the 2dF Galaxy Redshift Survey [31]. They appear as a distinct peak in the two-point correlation function at a scale of approximately 150 Mpc, corresponding to the sound horizon at recombination (see Fig. 1.7).

Additionally, dark matter behaved differently, as it does not interact with photons. Consequently, dark matter is affected only by gravity, creating overdensities that will be the foundation of Large Scale Structures 1.2.5.

1.2.4 Recombination

Recombination refers to the epoch when the Universe became sufficiently cool and dilute that photons could no longer efficiently maintain the coupling between matter and radiation, leading to the decoupling of the primordial plasma and the transparency of the universe. From this moment on, photons propagate freely through space, forming a relic radiation known as the *Cosmic Microwave Background* (CMB), discussed in detail in Section 1.3. During this transition, nuclei were no longer kept ionised by high-energy photons, allowing the formation of the first neutral atoms, mainly hydrogen and helium (see Section 1.2.2).

Prior to recombination, matter and radiation were tightly coupled, and the primordial plasma was in thermal equilibrium. As a consequence, the CMB is expected to exhibit an almost perfect blackbody spectrum, described

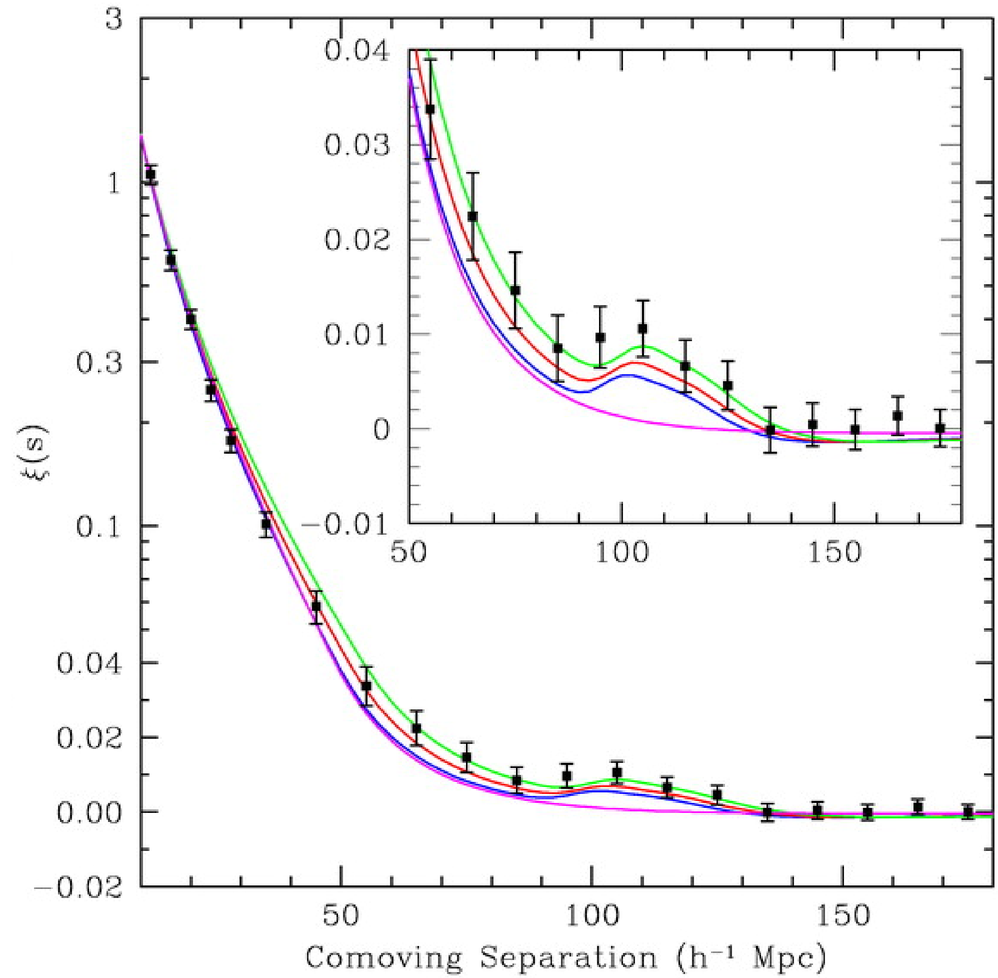

Figure 1.7: Matter two-point correlation function over comoving distance of the SDSS LRG sample. The BAO peak is visible at 150 Mpc, corresponding to an overdensity of matter at this distance. From [30].

by Planck's law [32]:

$$B_\nu(T) = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}. \tag{1.3}$$

This prediction has been confirmed by the COBE satellite, which provided the most precise measurement of a blackbody spectrum to date, as shown in Figure 1.8.

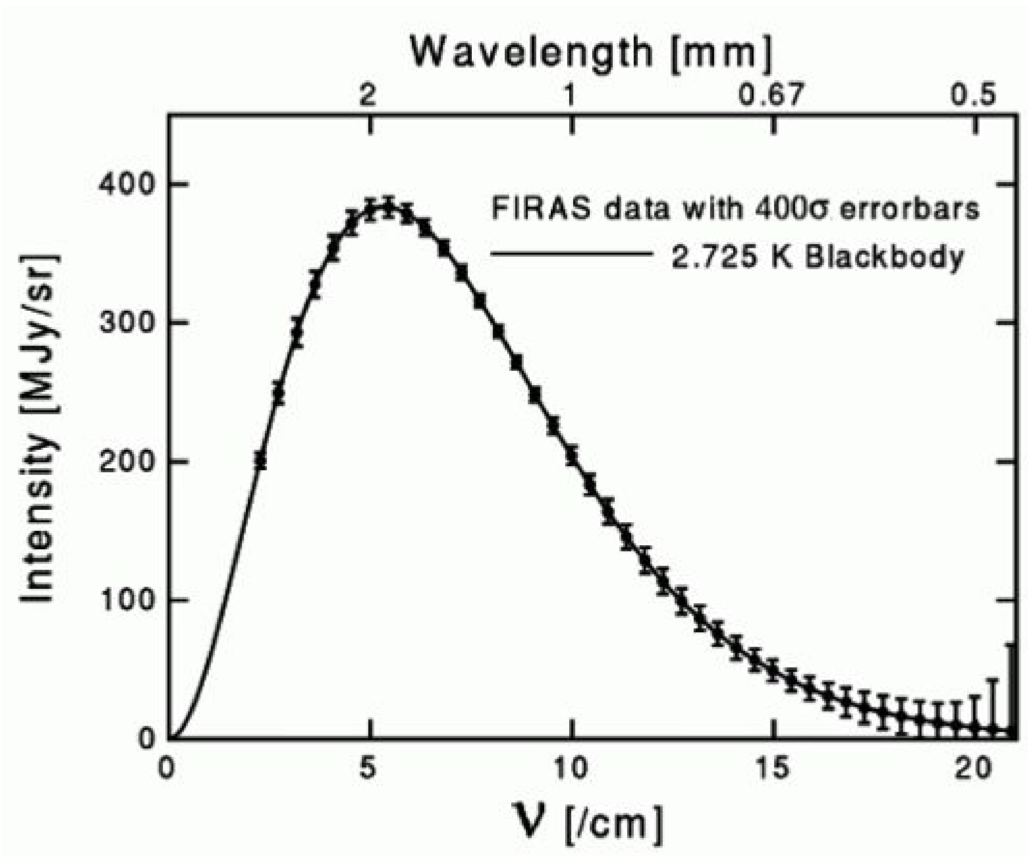

Figure 1.8: CMB spectrum measured by FIRAS instrument on COBE satellite. Plain line is the theoretical spectrum for a blackbody at 2.725 K, points are data from FIRAS with errorbars multiplied by 400 to be visible.

1.2.5 Large Scale Structures

After matter–radiation decoupling, baryonic matter evolves primarily under the influence of gravity. The small primordial density perturbations grow through gravitational instability. In the linear regime, their growth is proportional to the scale factor during matter domination. Dark matter, having decoupled earlier, forms gravitational potential wells into which baryons subsequently fall, leading to the formation of structures. This hierarchical process results in the emergence of the large-scale structure of the Universe, including halos, galaxies, clusters, and filaments. On large scales, the statistical properties of matter fluctuations are described by the matter power spectrum $P(k)$, which encodes both the primordial fluctuations and their subsequent evolution through cosmic expansion and gravitational collapse. A key observable imprint of these processes is given by Baryon Acoustic Oscillations (BAO), which provide a standard ruler in the distribution of matter (see Section 1.2.3).

1.3 Cosmic Microwave Background

So far, we have repeatedly referred to the Cosmic Microwave Background (see 1.2.4). However, as the CMB constitutes our primary observational probe of the primordial Universe and lies at the core of this thesis, we now present a more detailed discussion of its physical origin, its specific properties, and its connection to cosmological parameters, both through its intensity and polarization signals.

1.3.1 Angular Power Spectrum

Before discussing the physical details of the CMB, it might be relevant to intriduce introduce the mathematical tool to study the statistical properties of anisotropies distribution across the sky, namely the angular power spectrum. The angular power spectrum is a decomposition of a sky map into spherical harmonics, which describes the variance of the anisotropies as a function of the angular scale on the surface of a sphere. It is equivalent to a two-points correlation function in real space, but can apply only on a random Gaussian field.

Spherical Harmonics

Spherical harmonics are the eigenfunctions of the angular part of the Laplace operator on the sphere:

$$[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}]Y_{\ell m}(\theta,\phi) = -\ell(\ell+1)Y_{\ell m}(\theta,\phi). \tag{1.4}$$

They form a complete orthonormal basis on the sphere, which makes them particularly well suited to decompose any signal defined on the sky, such as CMB temperature anisotropies. The spherical harmonic functions $Y_{\ell m}(\theta,\phi)$ depend on the angular coordinates θ and ϕ, as well as on two integers: the multipole $\ell \in \mathbb{N}$, which characterises the angular scale on the sphere, and the azimuthal number m, such that $-\ell \leq m \leq \ell$. For each multipole ℓ, there are $2\ell+1$ independent modes. The angular scale associated with a given multipole is roughly given by $\theta \sim \pi/\ell$. The spherical harmonics can be written explicitly as:

$$Y_{\ell m}(\theta,\phi) = \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}P_\ell^m(\cos\theta)\exp^{im\phi}, \tag{1.5}$$

where P_ℓ^m are the associated Legendre polynomials (see e.g. [33] for more details).

Angular Power Spectrum

After introducing spherical harmonics, the temperature anisotropies (or any scalar field on the sphere) can be expanded as :

$$\Delta T(\vec{n}) = \sum_{\ell=0}^{+\infty}\sum_{m=-\ell}^{\ell} a_{\ell m}Y_{\ell m}(\vec{n}). \tag{1.6}$$

Using the orthogonality of spherical harmonics, the coefficients $a_{\ell m}$ are given by :

$$a_{\ell m} = \int_{4\pi}\Delta T(\vec{n})Y^*_{\ell m}(\vec{n})d\Omega \tag{1.7}$$

$$\approx \Omega_{pix}\sum_{p=1}^{N_{pix}}\Delta T(p)Y^*_{\ell m}(p), \tag{1.8}$$

where the second line corresponds to a discretized sky map.

The statistical properties of the field are fully described by the two-point correlation function. In harmonic space, this implies :

$$\langle a^*_{\ell m} a_{\ell\prime m\prime} \rangle = \delta_{\ell\ell\prime}\delta_{mm\prime}C_{ell}, \tag{1.9}$$

with $\langle\rangle$ designating average over many sky realisations, and C_ℓ is the angular power spectrum.

If the anisotropies are Gaussian, the coefficients $a_{\ell m}$ are independent random variables with zero mean and variance C_ℓ. An unbiased estimator of the angular power spectrum is then given by:

$$\hat{C}_\ell = \frac{1}{2\ell+1}\sum_{m=-\ell}^{\ell} |a_{\ell m}|^2, \tag{1.10}$$

which satisfies :

$$\langle \hat{C}_\ell \rangle = C_\ell. \tag{1.11}$$

Cosmic variance

Unfortunately, we only have access to one realisation of the sky, and independent modes for each multipole ℓ. Meaning that there is an intrinsic statistical uncertainty on the estimator $\hat{C}_\ell$, which cannot be reduced by improving instrumental sensitivity. This fundamental limitation is known as *cosmic variance*. The variance of the estimator $\hat{C}_\ell$ can be computed as :

$$\frac{\mathrm{Var}(\hat{C}_\ell)}{C_\ell^2} = \frac{\langle(\hat{C}_\ell - C_\ell)^2\rangle}{C_\ell^2} \tag{1.12}$$

$$= \frac{1}{(2\ell+1)^2 C_\ell^2}\left\langle \sum_{mm'} a^*_{\ell m} a_{\ell m}\, a^*_{\ell m'} a_{\ell m'} \right\rangle - 1 \tag{1.13}$$

$$= \frac{2}{2\ell+1}. \tag{1.14}$$

Sky pixellisation

In real observations, sky maps must be pixelized to allow for numerical treatment; consequently, the angular coordinates θ and ϕ are discretized. The most widely used pixelization scheme in CMB experiments (and cosmology in general) is the HEALPix method [34], which is accessible in the Python package Healpy[1]. The number of pixels in HEALPix, and thus the angular resolution, is controlled by the parameter N_{side}, with : $N_{\text{pix}} = 12N_{\text{side}}^2$. In principle, the maximum multipole that can be reliably probed depends on the pixel size. In practice, HEALPix relies on numerical approximations to accelerate spherical harmonic transforms, which become computationally expensive at high multipoles. To ensure accuracy, the maximum multipole is typically limited to $\ell_{\text{max}} = 3N_{\text{side}}$.

This framework is based on three key properties:

1. **Hierarchical structure.** Pixels are organized in a tree structure: increasing the resolution corresponds to subdividing each pixel into four sub-pixels. This allows for efficient multi-resolution analysis and facilitates fast local operations such as convolutions. Moreover, the indexing scheme preserves locality, meaning that nearby points on the sphere have nearby indices, which enables efficient neighbour searches.

2. **Equal area.** All pixels have exactly the same area, $\Omega_{\text{pix}} = \frac{4\pi}{N_{\text{pix}}}$, which ensures that white noise remains white in pixel space and avoids introducing spatial biases in the estimation of C_ℓ. It also simplifies numerical integrations on the sphere: $\int f(\hat{n})\, d\Omega \approx \Omega_{\text{pix}} \sum_p f_p$.

3. **Iso-latitude.** The centers of the pixels are arranged on rings of constant latitude, each containing pixels uniformly distributed in ϕ. This property significantly accelerates the computation of the spherical harmonic coefficients $a_{\ell m}$, since the associated Legendre polynomials $P_\ell^m(\cos\theta)$ only need to be evaluated once per ring. As a result, the computational complexity of spherical harmonic transforms is reduced from $\mathcal{O}(N_{\text{pix}}\ell_{\text{max}}^2)$ to $\mathcal{O}(N_{\text{pix}}^{1/2}\ell_{\text{max}}^2)$.

Beyond pixelization, HEALPix provides a comprehensive set of tools for spherical data analysis, including efficient algorithms for spherical harmonic transforms and angular power spectrum estimation, which are widely used in CMB studies and throughout this thesis.

[1] `https://github.com/healpy/healpy`

1.3.2 CMB Temperature

As discussed above (see 1.2.4), the CMB follows a nearly perfect blackbody spectrum. Its emission is therefore entirely determined by a single parameter: its temperature. Since the temperature of the photon–baryon plasma at recombination depends on local physical conditions, the observed CMB temperature anisotropies directly trace spatial fluctuations in this plasma. For this reason, CMB intensity anisotropies are conventionally expressed in terms of an effective temperature, which encodes information about the density fluctuations of the primordial plasma and, more generally, about its underlying physics.

Temperature Monopole and Dipole

The temperature monopole of the CMB is its mean temperature across the sky. It has been first measured by FIRAS instrument of COBE satellite at $T_0 = 2.7255 \pm 0.0006K$ [19]. This measurement remains the most precise determination of the CMB monopole, as subsequent experiments have primarily focused on measuring anisotropies around this mean temperature. From particle physcis considerations, photon–baryon decoupling occurs when the primordial plasma reaches a temperature $T_{rec} \sim 3000K$. As discussed in 1.1.1, cosmic expansion induces a redshift of photon wavelengths. For a blackbody spectrum, this implies that the temperature scales with redshift as :

$$T(z) = T_0(1 + z), \tag{1.15}$$

where T_0 is the present CMB temperature and z its redshift. Using the measured value of T_0 and the computed recombination temperature T_{rec}, one can infer that the CMB was emitted at a redshift $z_{\mathrm{rec}} \sim 1100$.

A first level of anisotropy is induced by the Earth's motion with respect to the CMB rest frame. Indeed, because the Earth has a peculiar velocity relative to the CMB rest frame, arising from the motion of the Solar System and of the Milky Way, the CMB photon wavelengths (and thus the observed temperature) are Doppler shifted due to the electromagnetic Doppler effect [3]. The CMB solar dipole was first observed by the DMR instrument aboard the COBE satellite [35], and later measured more accurately by WMAP [36] and Planck [37]. The solar dipole, simulated using Python Sky Model[2] (PySM)

[2] `https://github.com/galsci/pysm`

is shown in Figure 1.9.

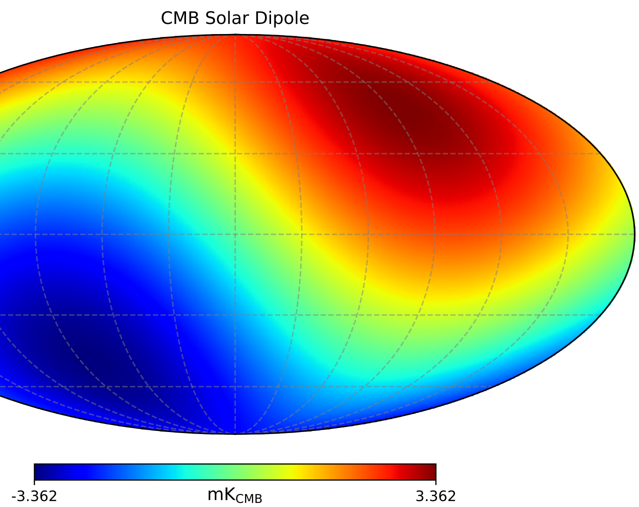

Figure 1.9: CMB solar dipole simulated using the PySM software [38] and calibrated using Planck 2018 measurements [37].

CMB Anisotropies

As the primordial plasma exhibits small density inhomogeneities at the level of $\sim 10^{-5}$, these are expected to manifest as temperature anisotropies in the Cosmic Microwave Background (CMB). The first detection of these anisotropies was made by the COBE satellite using its DMR instrument [35]. The full-sky maps obtained after four years of observation are shown in Figure 1.10.

These measurements were significantly improved by later experiments, in particular the Planck satellite, which achieved higher sensitivity and angular resolution in CMB temperature maps. The 150 GHz full-sky map is shown in Figure 1.11, taken from the Planck Legacy Archive[3].

[3] `https://pla.esac.esa.int/`

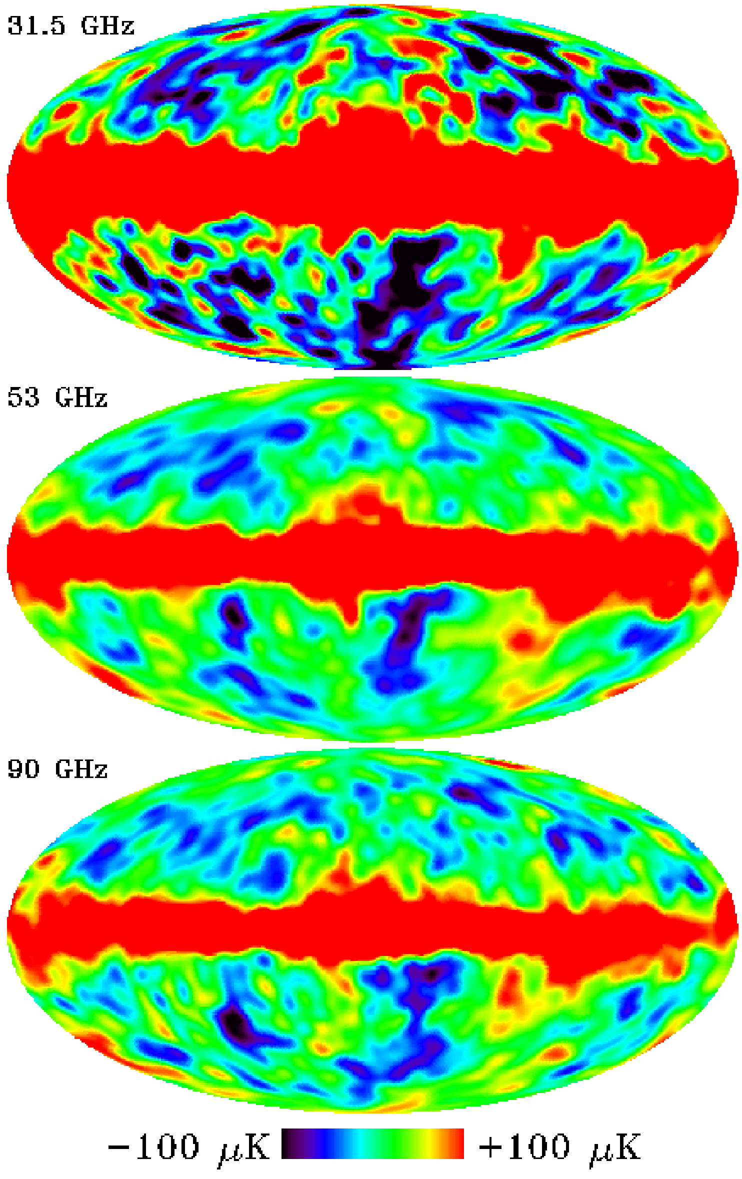

Figure 1.10: Full-sky maps from the DMR instrument aboard the COBE satellite after four years of observation at 31.5, 53 and 90 GHz. The monopole and dipole components have been removed. The bright band along the Galactic plane is due to Galactic emission, while cosmological temperature anisotropies are visible at high latitudes. Maps taken from [39].

The Planck satellite reached a regime in which the measurement of the temperature power spectrum is cosmic-variance limited over a wide range of angular scales. In this regime, the dominant source of uncertainty is no longer instrumental noise, but the fundamental statistical variance associated with observing a single realisation of the Universe. However, this limitation does not apply uniformly: at high multipoles, instrumental noise and foreground contamination become significant, while in polarization—especially B-modes—cosmic-variance-limited performance has not been reached.

The new generation of ground-based CMB experiments aims to improve measurements at small angular scales (high multipoles), as discussed in Section **??**.

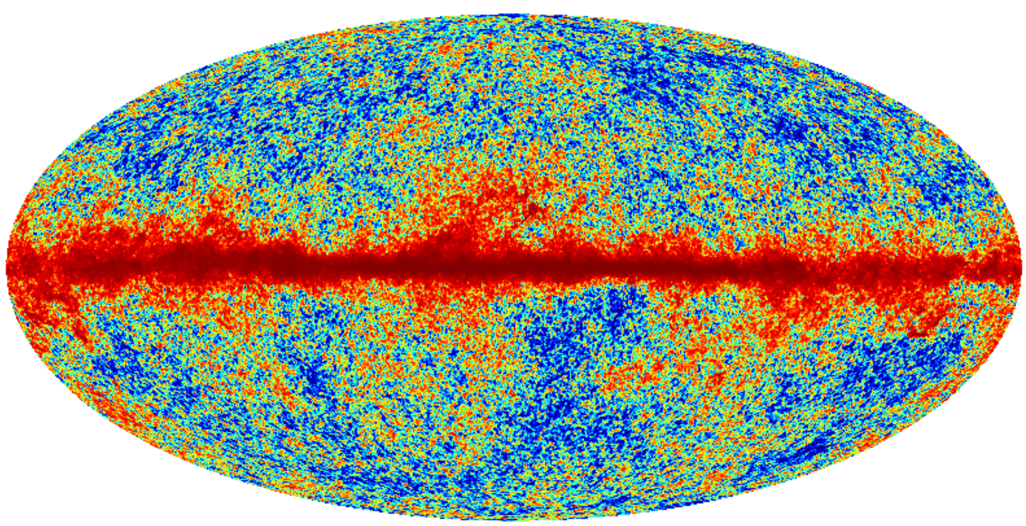

Figure 1.11: Full-sky CMB temperature map from the Planck satellite at 150 GHz. The monopole and dipole components have been removed. Compared to COBE, the improvement in angular resolution is clearly visible.

CMB anisotropy measurements are a key probe to understand the physics of the primordial plasma and, more generally, the mechanisms of the early Universe.

CMB power spectrum

The CMB anisotropies are expected to follow nearly Gaussian statistics, as expected by Inflation [40]. Observations by the Planck satellite have found no significant evidence for primordial non-Gaussianity in the temperature fluctuations. This justifies the use of the angular power spectrum as a complete statistical descriptor of the CMB anisotropies. From the Planck maps, the angular power spectrum shown in Figure 1.12 can be reconstructed. The large uncertainties at low multipoles (ℓ), corresponding to large angular scales, arise primarily from cosmic variance, as discussed previously. They are further exacerbated by observational limitations: emission from the Milky Way obscures part of the sky, preventing full-sky measurements and limiting access to the lowest multipoles of the CMB power spectrum. One can note that y-axis is showing D_ℓ instead of C_ℓ, D_ℓ only being a re-scaling of the angular power spectrum, as it represents the contribution to the variance per logarithmic interval in multipole ℓ. In particular, a scale-invariant primordial spectrum corresponds to a nearly constant D_ℓ at large angular scales, which is given by :

$$D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell. \tag{1.16}$$

The CMB angular power spectrum exhibits three main regimes, which we describe below.

Sachs-Wolfe effect : $\ell \lesssim 30$. At large angular scales, temperature anisotropies are dominated by gravitational redshift effects: photons climbing out of potential wells lose energy. This produces a nearly flat plateau in D_ℓ at low multipoles. The measurements in this regime are limited by cosmic variance and partial sky coverage [41].

Baryion Acoustic Oscillations : $30 \lesssim \ell \lesssim 2000$. The peaks in the CMB power spectrum arise from acoustic oscillations (see 1.2.3) in the tightly coupled photon-baryon plasma prior to recombination. The first peak, at $\ell \sim 220$, corresponds to the mode that has undergone half an oscillation at the time of last scattering. Subsequent peaks correspond to higher-order oscillations: odd peaks correspond to compressions in gravitational potentials, while even peaks correspond to rarefactions driven by photon pressure.

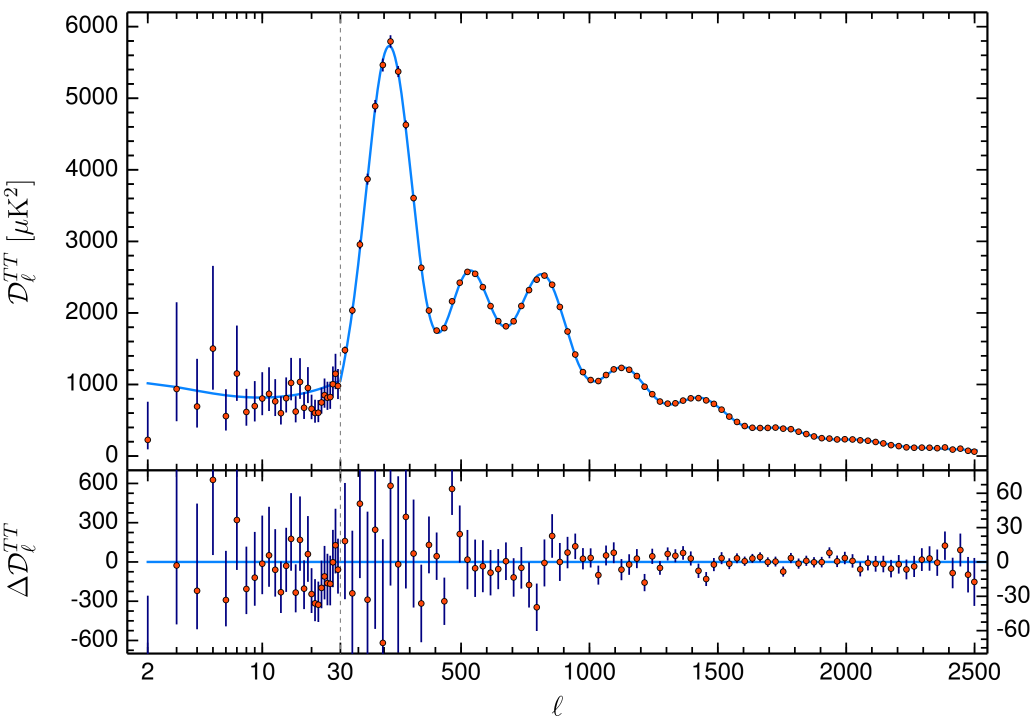

Figure 1.12: (Top) Best fit of the Angular power spectrum from Planck Temperature maps according to the minimal ΛCDM model. (Bottom) shows the difference between power spectrum computed from data and the best fit. Taken from [12].

Silk Damping : $\ell \gtrsim 1000-2000$. At small angular scales, anisotropies are suppressed due to photon diffusion (Silk damping) and the finite thickness of the last scattering surface. Photons undergo a random walk before decoupling, which erases small-scale temperature fluctuations. This leads to an exponential suppression of power at high multipoles [42].

1.3.3 CMB Polarisation

So far, we discussed the temperature anisotropies of the Cosmic Microwave Background. In addition, primordial mechanisms are also polarising CMB signal, up to 10% of the temperature amplitude signal. Studying the polarisation part of the signal provide rich information on primordial universe physics. We describe here these associated mechanisms and their implications on CMB signal.

Reminder on polarisation

Polarisation of the CMB plays a central role in this thesis. We briefly introduce here the relevant definitions and properties used in the following. Photons are described as electromagnetic waves, consisting of electric and magnetic fields governed by Maxwell's equations [43]. In Cartesian coordinates (x, y, z), an electric field propagating along the z-axis can be written as :

$$\vec{E}(t) = \begin{pmatrix} A_x e^{i(\omega t+\phi_x)} \\ A_y e^{i(\omega t+\phi_y)} \\ 0 \end{pmatrix}, \tag{1.17}$$

where A_x and A_y are the amplitudes, ω is the frequency, and ϕ_x, ϕ_y are the phases.

Stokes parameters The polarisation state of the radiation is fully described by the Stokes parameters [44], which can be defined through the coherence matrix:

$$C = \begin{pmatrix} \langle |E_x|^2 \rangle & \langle E_x E_y^* \rangle \\ \langle E_x^* E_y \rangle & \langle |E_y|^2 \rangle \end{pmatrix} \tag{1.18}$$

$$= \frac{1}{2} \begin{pmatrix} I+Q & U-iV \\ U+iV & I-Q \end{pmatrix}. \tag{1.19}$$

From this definition, the Stokes parameters are given by:

$$I = \langle |E_x|^2 \rangle + \langle |E_y|^2 \rangle, \tag{1.20}$$

$$Q = \langle |E_x|^2 \rangle - \langle |E_y|^2 \rangle, \tag{1.21}$$

$$U = 2\,\mathrm{Re}\left(\langle E_x E_y^* \rangle \right), \tag{1.22}$$

$$V = 2\,\mathrm{Im}\left(\langle E_x E_y^* \rangle \right). \tag{1.23}$$

The parameter I corresponds to the total intensity of the radiation. The parameters Q and U describe linear polarisation, while V characterises circular polarisation. In the context of the CMB, no significant circular polarisation has been detected, and V is therefore neglected.

A visual representation of the Stokes parameters is shown in Figure 1.13.

E- and B-modes The Stokes parameters Q and U depend on the choice of local coordinate system and are therefore not invariant under rotations on the sphere. Under a rotation by an angle ψ, they transform as :

$$(Q \pm iU)'(\hat{n}) = e^{\mp 2i\psi}(Q \pm iU)(\hat{n}), \tag{1.24}$$

which shows that they are spin-2 quantities.

To construct rotationally invariant quantities, [**zaldarriaga1997**] introduced a decomposition in terms of spin-weighted spherical harmonics:

$$(Q \pm iU)(\hat{n}) = \sum_{\ell m} a_{\ell m}^{\pm 2}\, {}_{\pm 2}Y_{\ell m}(\hat{n}). \tag{1.25}$$

From these coefficients, the scalar E-mode and pseudo-scalar B-mode components are defined as :

$$a_{\ell m}^{E} = -\frac{1}{2}\left(a_{\ell m}^{2} + a_{\ell m}^{-2} \right), \tag{1.26}$$

$$a_{\ell m}^{B} = \frac{i}{2}\left(a_{\ell m}^{2} - a_{\ell m}^{-2} \right). \tag{1.27}$$

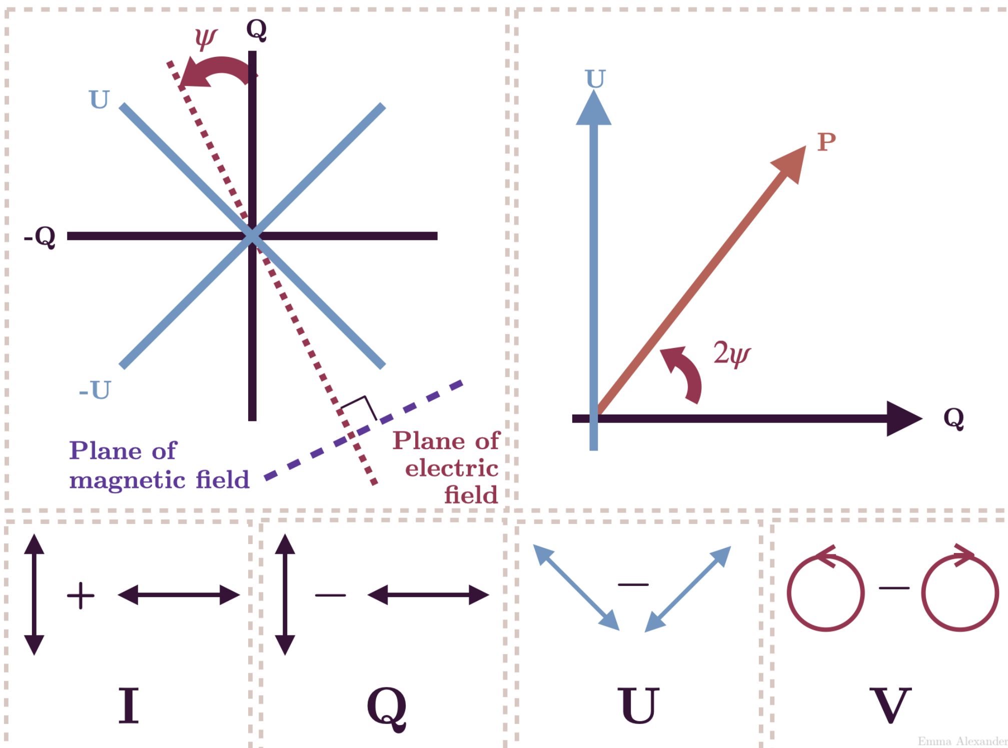

Figure 1.13: Visual representation of the Stokes parameters. Image from Wikimedia Commons [45].

The corresponding real-space fields are then :

$$E(\hat{n}) = \sum_{\ell m} a_{\ell m}^{E} Y_{\ell m}(\hat{n}), \tag{1.28}$$

$$B(\hat{n}) = \sum_{\ell m} a_{\ell m}^{B} Y_{\ell m}(\hat{n}). \tag{1.29}$$

E-modes correspond to gradient-like patterns and are even under parity transformations. They are primarily generated by scalar perturbations. In contrast, B-modes correspond to curl-like patterns, are odd under parity, and can be generated by tensor perturbations (primordial gravitational waves) as well as by gravitational lensing[4].

A visual representation of the E- and B-mode patterns is shown in Figure 1.14.

Compton Scattering

Gravitational Waves

1.3.4 Cosmological parameters

The minimal ΛCDM model is described by six free cosmological parameters. Although the model can be extended to include additional parameters, such extensions often introduce degeneracies and are not required to provide a significantly better fit to CMB observations [12]. The six parameters of the ΛCDM model are:

- **Baryon density** $\Omega_b h^2$,
- **Cold dark matter density** $\Omega_c h^2$,
- **Angular size of the sound horizon at recombination** θ_{MC},
- **Optical depth to reionization**[5] τ,
- **Amplitude of scalar fluctuations** $\ln(10^{10} A_s)$,

[4] Their name was given by analogy to the electric and magnetic fields' properties.

[5] Reionization occurred after recombination, when the first luminous sources formed and emitted high-energy photons that progressively ionized the neutral hydrogen in the intergalactic medium.

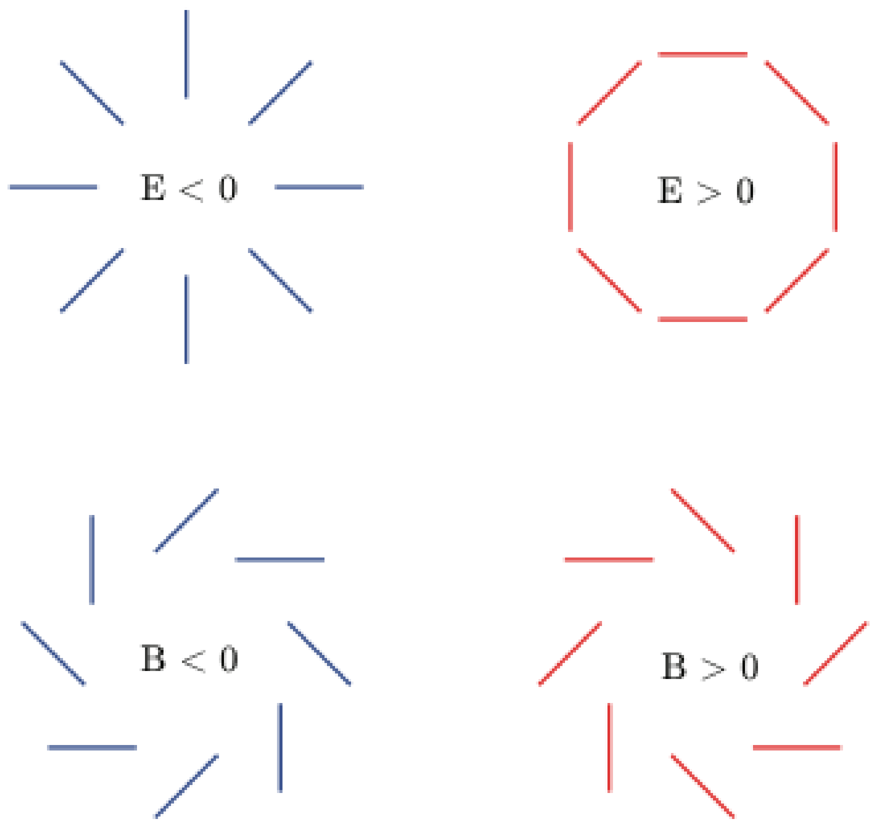

Figure 1.14: Visual representation of the E and B modes. Image from Wikimedia Commons [46].

- **Spectral index of scalar fluctuations** n_s.

In order to constrain these parameters, additional assumptions are typically made. The number of relativistic neutrino species is fixed to $N_{\text{eff}} = 3.046$, and the sum of neutrino masses is assumed to be $\sum m_\nu = 0.06\,\text{eV}$. The Universe is assumed to be spatially flat ($\Omega_k = 0$), and dark energy is modeled as a cosmological constant ($w = -1$).

The resulting constraints on these parameters from the Planck 2018 data, combining temperature and polarization power spectra, are shown in Figure **??**. From these fitted values, other cosmological parameters can be derived, such as the total matter density Ω_m, the Hubble parameter H_0, and the age of the Universe t_0.

Should I go further ?

1.4 CMB contaminations

1.4.1 Lensing

Should I add lensing in the previous section ? As it is an additional source of polarisation... But, as it is a non-primordial source of polarisation, I believe that it might be in CMB contamination.

1.4.2 Astrophysical Dust

1.4.3 Synchrotron

1.4.4 Atmosphere

1.5 Past & Future Experiments

1.5.1 Past Experiements

1.5.2 Current Experiments

1.5.3 Future Experiments

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